

EMPIRICAL RESEARCH · BITCOIN FACTORS · CORRELATION VS. LEADING INDICATORS

What actually drives Bitcoin: a high-beta risk asset that follows global liquidity — not digital gold, and not a falling-dollar trade.

v1.2 · Sample 2015–2026 (power law & production floor from 2010) · first-party empirical engine + STORM cross-check ·

+29.8pp

BTC FWD-6M RETURN SPREAD
WHEN M2 IS ACCELERATING
(THE TOP LEADING INDICATOR)

0.28 → 0.52

BTC-NASDAQ DAILY
CORRELATION: FULL SAMPLE
VS. LAST 12 MONTHS (RISK-
BETA, RISING)

+0.09

BTC-GOLD FULL-SAMPLE
DAILY CORRELATION —
STATISTICALLY ~ZERO
("DIGITAL GOLD" FAILS)

Executive summary

The data overturns the lazy version of every Bitcoin narrative. Bitcoin is, first and foremost, a **high-beta risk asset whose equity correlation has structurally risen** — its daily correlation to the Nasdaq climbed from ~0.28 over the full 2015–26 sample to ~0.52 over the last year — not "digital gold" (full-sample gold correlation ~0.09, statistically indistinguishable from zero). The single most useful *leading* indicator is **global liquidity / M2 growth**: when US M2 year-over-year is accelerating, BTC's median forward-6-month return is +35.0% versus +5.2% when decelerating — a +29.8pp spread. Crucially, the factors most people *watch* — the Nasdaq, gold, the VIX, credit spreads — are **coincident correlates with no predictive lead**, not leading indicators. And the popular "weak-dollar / falling-real-rates" trade is the biggest myth: on a forward-return basis it is perverse in-sample. This note separates what moves *with* Bitcoin from what moves *before* it, and grades six canonical assumptions against the evidence. It is analysis, not advice, and contains no recommendation regarding any security or digital asset.

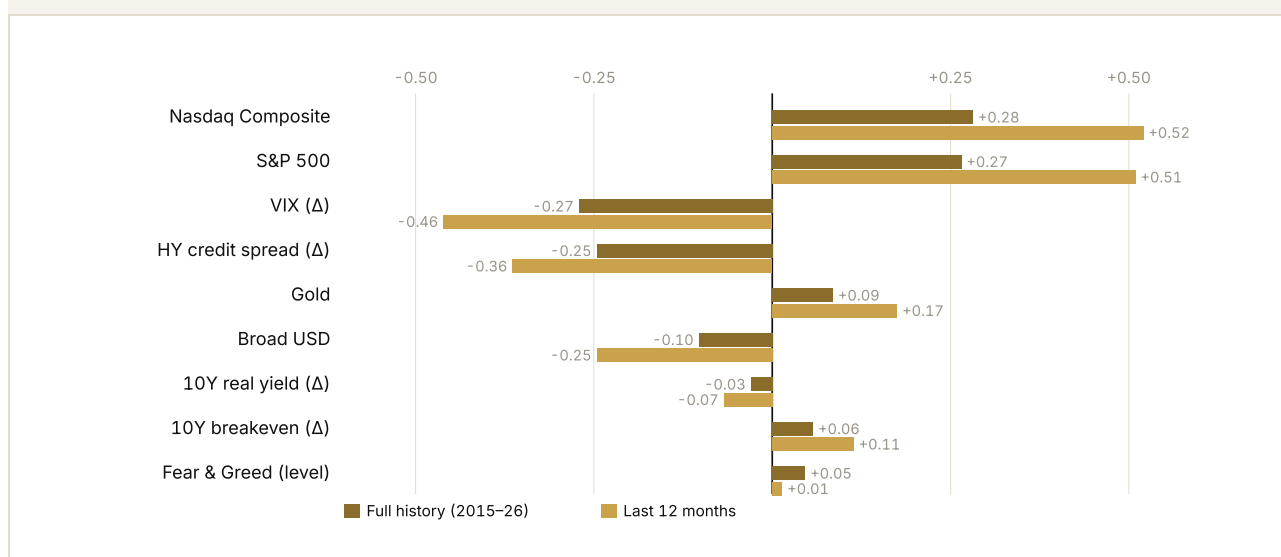
The distinction that organizes this note. *Contemporaneous correlation* is co-movement at the same time (coincident beta) — useful for hedging and risk, useless for timing. A *leading indicator* has predictive power with a lead (factor today → BTC return later) — that is what an allocator can act on. Conflating the two is the most common analytical

error in this field: a factor can be highly correlated yet have zero predictive value (the Nasdaq), or weakly correlated day-to-day yet a strong forward signal (M2).

1 · Contemporaneous correlation — and its instability

Full-sample daily-return correlations are mostly modest; the story is in the **windows**. Most factors are near-zero over the full sample but strongly correlated over the last 12 months — Bitcoin's correlation structure has shifted hard toward risk-asset behavior in the post-spot-ETF era.

EXHIBIT 1 · DAILY-RETURN CORRELATION TO BTC — FULL HISTORY VS. LAST 12 MONTHS



Pearson correlation of daily log returns (yields/spreads/vol in first differences). Source: TON618 factor engine, 2026-06-18 17:17 UTC. M2 and Fed-balance-sheet contemporaneous figures are YoY autocorrelation artifacts and are assessed as leading indicators below, not here.

2 · Coincident vs. leading — the critical split

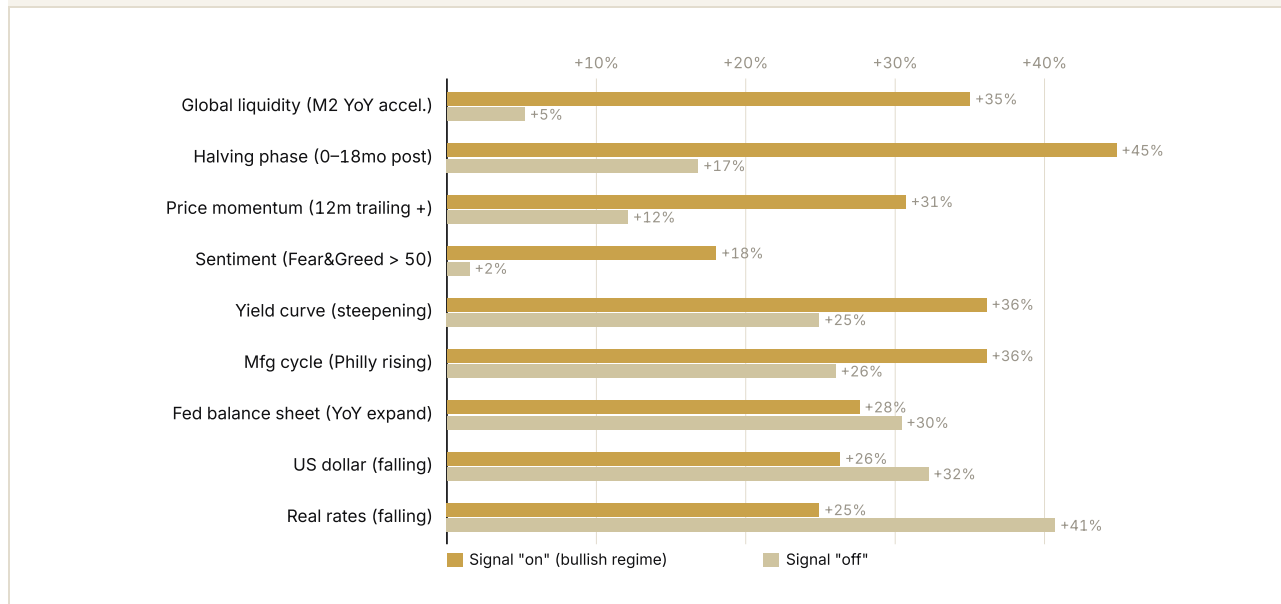
Lead/lag cross-correlation (the lag at which each factor's correlation to BTC peaks) delivers the report's sharpest result:

- **Coincident — correlates, NOT predictors** (peak at lag 0): Nasdaq, S&P 500, VIX, HY credit spreads, gold, the broad dollar. *Everything the market watches as a "Bitcoin tell" moves with Bitcoin, not before it.*
- **Lagging** (Bitcoin leads them): Treasury yields, real yields, transaction count.
- **Leading — genuine forward edge**: global liquidity / M2, Bitcoin's own price momentum, the macro growth-vs-inflation regime, and the sentiment regime. These are few, and they are the only factors with timing value.

3 · Leading-indicator ranking — forward-6-month regime split

Median BTC forward-6-month return when each signal is "on" (bullish regime) versus "off," monthly 2015+:

EXHIBIT 2 · MEDIAN BTC FORWARD-6-MONTH RETURN BY SIGNAL STATE



Source: TON618 factor engine + in-house halving/momentum/PMI models, 2026-06-18 17:17 UTC. Bitcoin's own 12-month price momentum is one of the strongest leading signals (+18.6pp). The dollar and real-rate signals are the **wrong sign** versus popular belief — in-sample, BTC delivered higher forward returns when the dollar was firm and real rates were rising. Fed-balance-sheet YoY and the manufacturing cycle (Philly Fed, +10pp) carry little-to-no timing edge.

4 · Momentum: Bitcoin trends

Time-series momentum — Bitcoin's own trailing return as a predictor of its forward return — is a genuine, robust leading factor that the "macro-driver" debate usually ignores. When the trailing 12-month return is positive, the median forward-6-month return is **+30.7%** versus +12.1% when negative (a +18.6pp spread, 66% hit rate); the trailing-6-month → forward-3-month test shows +14.1% vs -1.9%. Bitcoin persists in its trend more often than not — a property it shares with other risk assets and commodities, and one that operates independently of any macro narrative.

5 · The macro regime that matters: growth × inflation

Conditioning Bitcoin's forward return on the four-quadrant (Hedgeye GIP) macro regime — growth accelerating/decelerating × inflation accelerating/decelerating — resolves the inflation-hedge question directly. Bitcoin's best regime is **Goldilocks** (growth accelerating, inflation *decelerating*); its worst is **Stagflation** (growth decelerating, inflation accelerating). An asset that does its best

when inflation is *falling* and its worst when inflation is *rising* is not an inflation hedge — it is a disinflationary-growth (risk-on) asset.

EXHIBIT 3 · BTC FORWARD-6-MONTH MEDIAN RETURN BY MACRO QUADRANT (GROWTH × INFLATION)

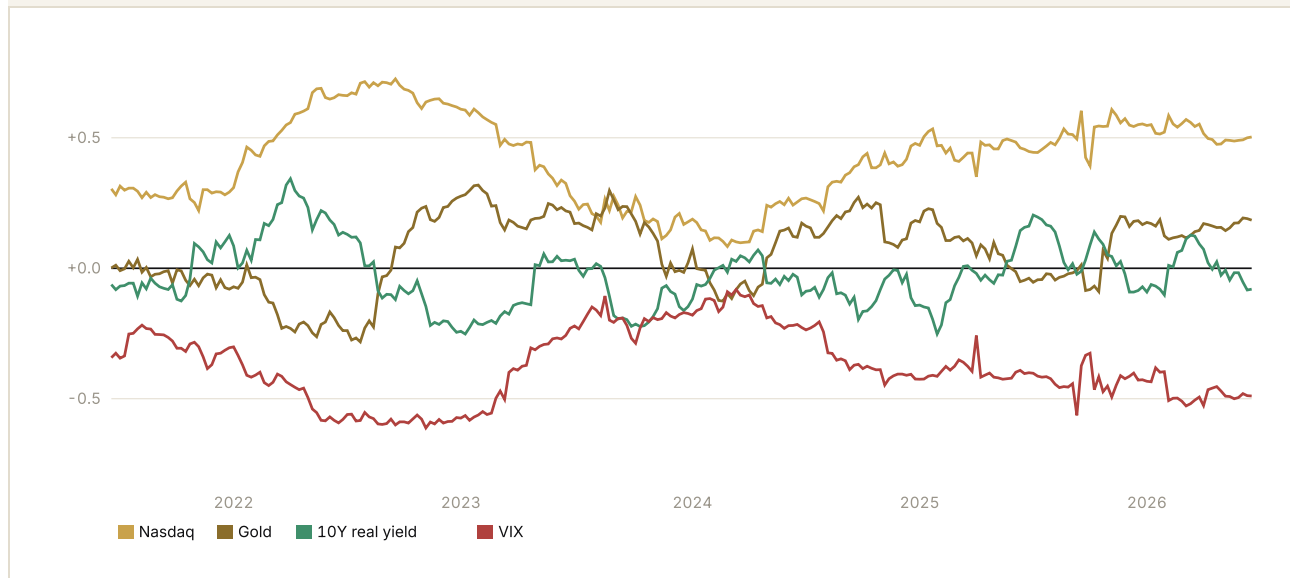


BTC median forward-6-month return by macro quadrant · darker = higher

Source: TON618 factor engine — growth = Philly Fed manufacturing diffusion (3-month acceleration; no free ISM series exists on FRED), inflation = CPI YoY (3-month acceleration), monthly 2015+. Quadrant labels follow the Hedgeye Growth-Inflation-Policy framework / All-Weather "seasons." 2026-06-18 17:17 UTC.

6 · No correlation is stable

EXHIBIT 4 · ROLLING 90-DAY CORRELATION OF DAILY RETURNS (5 YEARS)



Source: TON618 factor engine, 2026-06-18 17:17 UTC. BTC-Nasdaq has ratcheted toward +0.5; BTC-real-yield turned sharply negative only recently; BTC-gold is positive but unstable. The instability is itself the most robust finding — any single-number correlation is regime-conditional.

7 · Cycle position, valuation & structural floors

Three models a flat factor scan misses — each a stronger or more structural signal than any single macro correlation, and each one you flagged.

7.1 Manufacturing cycle — the turning points, not the level

PMI *level* alone is near-useless for Bitcoin (~0 contemporaneous correlation). Positioned in the business cycle, it is powerful. Bitcoin's best window is the **Recovery phase** — manufacturing below trend and turning up, just past a cycle *trough*: a +70.6% median forward-6-month return at a 88% hit rate. Its worst is the **Slowdown phase** — above trend and rolling over, just past a *peak*. The spread between catching the trough and chasing the peak is the entire point.

EXHIBIT 5 · BTC FORWARD-6-MONTH RETURN BY MANUFACTURING-CYCLE PHASE

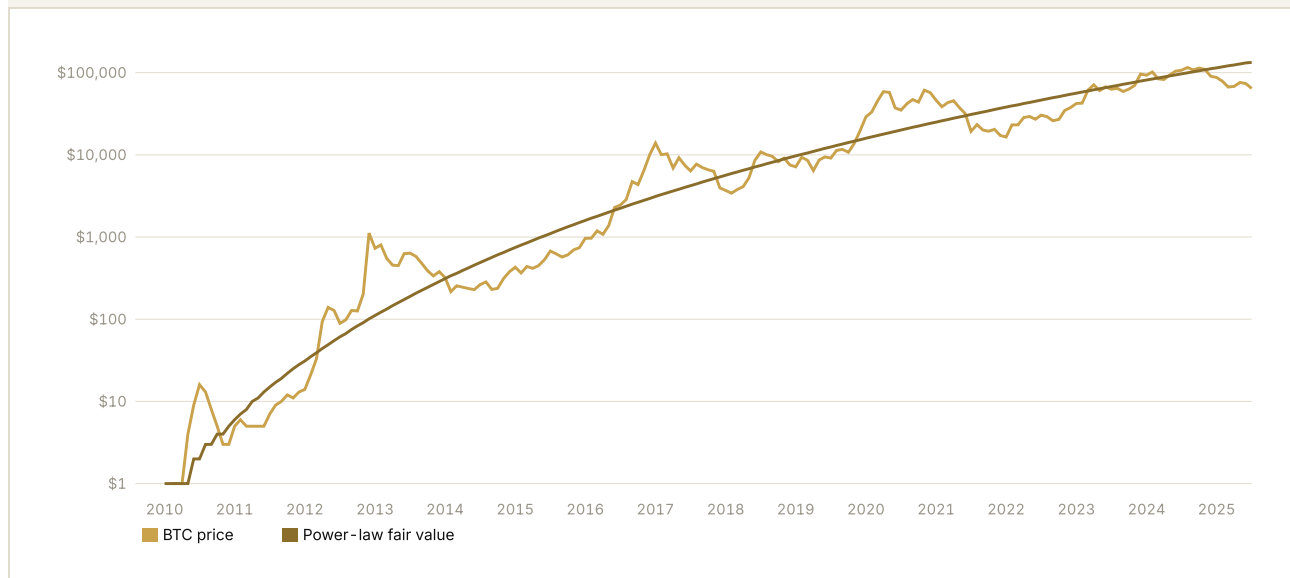
CYCLE PHASE	BTC FWD-6M (MEDIAN)	HIT	N
Recovery (below-trend & rising — post-trough)	+70.6%	88%	17
Expansion (above-trend & rising)	+40.7%	66%	70
Slowdown (above-trend & falling — post-peak)	+18.5%	53%	66
Contraction (below-trend & falling)	+43.3%	79%	33

Source: TON618 factor engine, Philly Fed manufacturing diffusion (3-month smoothed; centered at 0) as the ISM proxy — no free true ISM series exists on FRED. Phases = position (above/below trend) × direction (rising/falling). 2026-06-18 17:17 UTC.

7.2 Power law — Bitcoin's long-run trend and its deviation oscillator

Bitcoin's price has tracked a power law of time with remarkable fidelity: $\log \text{price} \approx a + 5.668 \cdot \log(\text{days since genesis})$, $R^2 = 0.9616$. The model itself is a trend, not a timing tool — but its *derivative*, the deviation of price from power-law fair value, is a valuation oscillator. When price trades **below** fair value, the median forward-6-month return is +70.6% (92% hit, $n=103$); when **richly extended** (top quintile of deviation), it is -24.9%. As of this writing Bitcoin sits about -52% versus a power-law fair value near \$133,530 — i.e., cheap on this model.

EXHIBIT 6 · BTC PRICE VS. POWER-LAW FAIR VALUE (LOG SCALE)

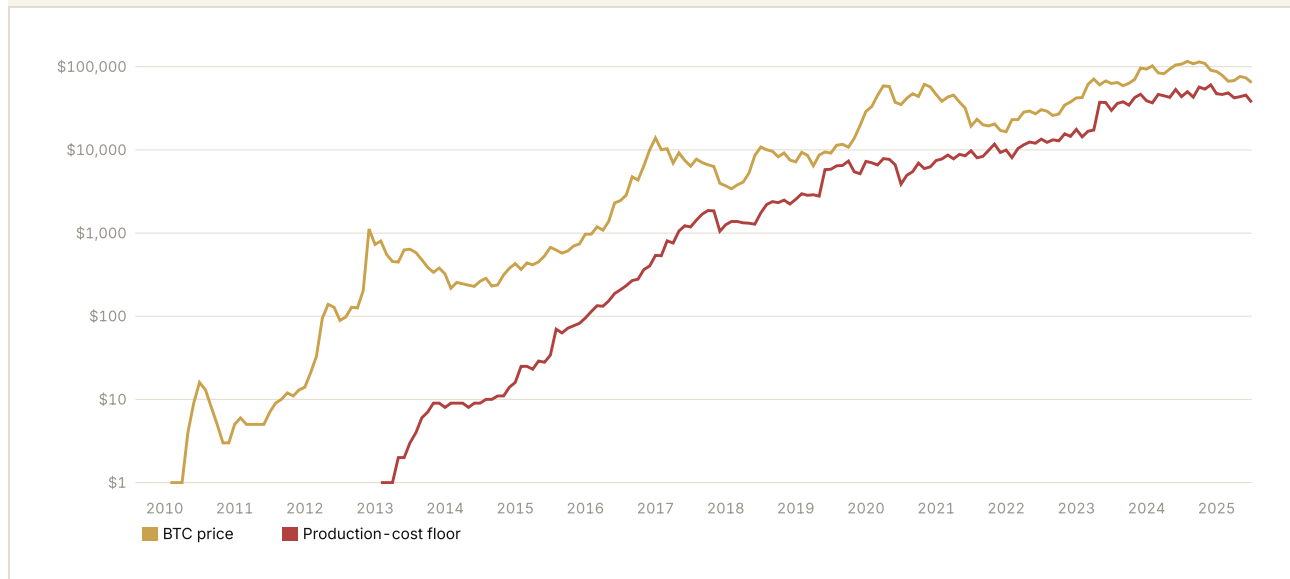


Source: TON618 factor engine; OLS fit of $\log_{10}(\text{price})$ on $\log_{10}(\text{days since the 2009-01-03 genesis})$, full history. Caveats: the exponent drifts as more data accrues; a high R^2 on a monotonic trend overstates predictive precision; the deviation oscillator is the actionable part, not the trend line.

7.3 Electricity / production-cost floor

Mining gives Bitcoin a soft price floor: when price approaches the marginal cost of production, high-cost miners capitulate, difficulty falls, and supply pressure eases. A transparent energy model puts current production cost near **\$37,238** — price is 1.73× that. The structural finding: across the full history price has **never closed below ~1.14× production cost** (0.0% of days below cost), and buying when price sits in the bottom quartile versus cost (near the floor) returned +38.5% over the next 6 months at a 79% hit rate.

EXHIBIT 7 · BTC PRICE VS. MODELED PRODUCTION-COST FLOOR (LOG SCALE)



Illustrative model — assumptions stated: fleet efficiency declining ~80→16 J/TH (exponential), electricity \$0.05/kWh, 144 blocks/day, block-reward schedule by halving. The level is approximate and sensitive to the electricity-price and efficiency assumptions; the robust result is that price has historically respected the floor, not the exact dollar figure. 2026-06-18 17:17 UTC.

8 · The assumption scorecard

Six canonical beliefs, graded against the first-party evidence:

COMMON ASSUMPTION	VERDICT	EMPIRICAL BASIS
"Bitcoin is digital gold / an inflation hedge"	Mostly FALSE	Full-sample BTC–gold daily corr +0.09 (≈ 0); gold is coincident; 10Y-breakeven corr ≈ 0 . By macro quadrant, BTC's <i>worst</i> regime is stagflation and its <i>best</i> is Goldilocks (inflation falling) — the inverse of an inflation hedge. A 2024–26 gold uptick to +0.31 (90-day) exists but is small.
"Bitcoin is a risk-on Nasdaq / tech-beta asset"	TRUE — strengthening	Daily corr 0.28 (full) $\rightarrow$ 0.52 (1y); VIX -0.46 , HY -0.36 over 1y; correlation rises in risk-off. The dominant contemporaneous relationship.
"Bitcoin is driven by global liquidity / M2, with a lead"	TRUE — best leading indicator	M2-accelerating regime $\rightarrow$ +35.0% vs +5.2% fwd-6m (+29.8pp, corr +0.33); in-house overlay shows ~ 80 -day lead. Note: Fed-balance-sheet YoY alone has no timing edge.
"The ~ 4 -year halving cycle drives price"	FRAGILE / partly true	0–18mo post-halving $\rightarrow$ +44.8% vs +16.8% fwd-6m (+28.1pp), but the forward-12-month spread collapses to +0.7pp and the test rests on ~ 3 independent cycles; heavily confounded with liquidity.
"Bitcoin is a weak-dollar / falling-real-rates debasement trade"	DENIED as a predictor	Forward-return spreads are perverse: dollar-falling -5.9 pp, real-rates-falling -15.8 pp. Only a weak, recent <i>coincident</i> dollar effect (-0.25 , 1y).
"On-chain metrics (MVRV/NUPL/SOPR) time tops & bottoms"	MIXED — useful at extremes	First-party valuation metrics auth-gated this run; network-usage on-chain is reflexive/coincident, not leading. Published backtests (3 cycles) credit MVRV/NUPL with real risk-adjusted edge at extremes — but as zones, on ~ 3 samples (see external corroboration).

External corroboration

Independent published evidence gathered via a STORM-method, perspective-structured research pass (Stanford OVAL method, adapted); each claim cited in Sources. Where external evidence and our first-party empirics agree, confidence rises; where they diverge, we flag it.

ASSUMPTION	WHAT THE PUBLISHED LITERATURE / PRACTITIONER EVIDENCE SHOWS
Digital gold / inflation hedge	Corroborated (fails). A factor-model study finds BTC's adjusted R^2 on equities jumped from ~0.11 pre-ETF to ~0.30 post-ETF, and the ETF-impact paper concludes managers should treat BTC as "a high-growth technology proxy rather than a defensive asset"; BTC's equity correlation peaked ~0.87 in 2024. ^{[3][4]} Our own four-quadrant test agrees: BTC's <i>worst</i> macro regime is stagflation (growth-/inflation+) and its <i>best</i> is Goldilocks (growth+/inflation-) — the inverse of an inflation hedge.
Risk-on Nasdaq / tech beta	Confirmed, with a nuance. The correlation <i>level</i> rose post-ETF (multiply corroborated). But adversarial verification flags that the same paper's Chow test on the <i>instantaneous beta</i> found <i>no</i> structural break ($p=0.87$) and its rolling-correlation Chow statistic (139–269) is autocorrelation-biased — so the co-movement level rose more cleanly than the beta did. ^[3] BTC also carries standalone downside tail risk: in the Oct-10-2025 leverage flush it fell ~30% while the Nasdaq barely moved. ^[8]
Global liquidity / M2 (with a lead)	Confirmed but time-varying. Time-varying Granger-causality work finds M2→BTC causality is sporadic and regime-dependent, with no full-sample cointegration on global M2; ^[5] a separate US-M2 study does find cointegration (long-run elasticity 2.65, CI 2.06–3.24). ^[6] Practitioners note BTC's M2 R^2 fell from 0.71–0.90 (2022) to ~0.59 by early 2026 as ETF flows became the dominant transmission channel ^[7] — matching our finding that M2 leads but the relationship is unstable.
On-chain valuation times tops & bottoms (MVRV/NUPL)	Partly corroborated, with caveats. A peer-reviewed backtest across three cycles (2013–2025) finds the MVRV Z-score beats buy-and-hold and Monte-Carlo random entry, flagging tops/bottoms — but as <i>zones</i> , on only ~3 independent cycles, with maturation risk. ^[9]
Weak-dollar / falling-real-rates debasement trade	Corroborated (denied as a tradeable signal). The headline BTC-dollar level correlation (~–0.58) is spuriously inflated by shared trends; on a tradeable <i>returns</i> basis it is ~0.04 same-day and only ~–0.20 at the best 33-day lead — weak, matching our perverse forward-return result. ^[10]

Methodology & data

Sample: daily, 2015-01-01 to 2026-06-17. **Primary, first-party data:** BTC price and on-chain metrics — CoinMetrics community API; macro, rates, credit, liquidity — FRED; gold and equity indices — Yahoo Finance; sentiment — alternative.me Fear & Greed. Correlation is computed on returns/changes, never raw trending levels (price → log return; yields/spreads/vol → first difference; slow liquidity → YoY). Estimators: Pearson and Spearman over four windows; lead/lag cross-correlation; rolling-90-day correlation; a forward-return regime split (median BTC fwd-6-month return, bullish vs bearish); a four-quadrant growth×inflation regime map; a time-series-momentum test; a manufacturing business-cycle phase model (position × direction, capturing troughs and peaks); a power-law fit (log price on log time-since-genesis) with its deviation oscillator; and a transparent production-cost / electricity-floor model. **Disclosed gaps:** on-chain valuation metrics (realized cap, MVRV) were auth-gated this run and are adjudicated via published evidence; no free true ISM PMI exists on FRED, so the Philly Fed diffusion index is used as the manufacturing-cycle proxy; the HY credit-spread series is available only from mid-2023; small-sample cycle claims (halving, $n \approx 3$) are flagged. Methods: in-house empirical engine (primary) plus

a STORM perspective - structured qualitative cross-check (Stanford OVAL method, adapted; 12 perspectives, 187 findings and 34 adversarial verdicts recovered and folded into the external-corroboration section after the run reached its compute limit during final synthesis).

Risk factors & limitations

- **Regime instability is the master risk** — every correlation is window-dependent and may invert; the post-2024 ETF era is a small sample.
- **Endogeneity** — on-chain network metrics are partly caused by price; not exogenous drivers.
- **Small-sample cycle claims** — the halving rests on ~3 independent cycles and is not statistically robust.
- **In-sample inference** — 2015–26 contains few independent macro cycles; forward returns are not guaranteed.

What we are watching

- Whether the post-ETF jump was in the correlation *level* or the *instantaneous beta* — the structural-break evidence conflicts (rolling-correlation Chow highly significant; beta Chow not), and a 2026 partial-decoupling has appeared.
- The M2 cointegration debate — a US-M2 study finds a stable 2.65 long-run elasticity, while global-M2 work finds no cointegration and only time-varying causality; reconcile scope (US vs global) and method.
- Whether the BTC price-momentum edge persists out-of-sample or decays as the asset institutionalizes.
- Power-law durability — does the ~5.7 exponent hold or decay as the asset matures (the model guarantees deceleration); and does the current ~50% -below-fair reading mean-revert as it has historically.
- Production-floor robustness — re-estimate with live fleet-efficiency and regional electricity data; watch whether the ~1.1× historical floor holds post-halving as issuance falls.
- A true ISM series (vs the Philly Fed proxy) for the business-cycle phase model, and replicating the Recovery-phase signal on global PMIs.
- MVRV / NUPL / SOPR cycle-timing efficacy — obtain a non-gated realized-cap source and test first-party.
- Whether the 2024–26 BTC–gold warming is durable, and whether the halving signal survives once liquidity is controlled for.

Sources

1. TON618 Capital factor engine (first-party): `factor_analysis.py`, run 2026-06-18 17:17 UTC.
Inputs: CoinMetrics community API (BTC price, on-chain); FRED / Federal Reserve Bank of St. Louis

- (macro, rates, credit, liquidity); Yahoo Finance (gold, equity indices); alternative.me (Fear & Greed).
2. TON618 Capital in-house models: `btc_halving_phase.py` (halving-cycle forward returns), `btc_m2_correlation.py` (M2 lead overlay), `btc_regime_ranking.py`.
 3. Hong, Feng, Wang & Li (Olin Business School, WUSTL), "Impact of Bitcoin ETF approval on hedging properties," arXiv 2512.12815 (2025-12) — incl. factor-model R^2 and the Chow-test caveat surfaced in adversarial review.
 4. Di Wu, "Institutional Adoption and Correlation Dynamics," arXiv 2501.09911 (2025-01) — BTC-equity correlation peaked ~ 0.87 in 2024.
 5. Gu & Chen, "Global M2 Money Supply and Bitcoin: A time-varying causality approach," Jönköping (2025) — sporadic, regime-dependent Granger causality; no full-sample cointegration on global M2.
 6. Kokabian, "The M2-Bitcoin Elasticity: A Cointegration Analysis (2015-2025)," Preprints.org 202506.1963 — US-M2 Johansen elasticity 2.65 [2.06-3.24].
 7. CF Benchmarks (G. Selby, 2026) — BTC's M2 R^2 fell 0.71-0.90 (2022) $\rightarrow \sim 0.59$ (Feb 2026); ETF flows the primary transmission channel since 2024.
 8. Crypto.com, BTC-Nasdaq Oct-2025 decoupling / Oct-10 leverage flush (BTC -30% on a $\sim \$19B$ liquidation cascade).
 9. On-chain measures backtest (NUPL, MVRV Z-score, CVDD), ScienceDirect S0275531926002138 (2013-2025).
 10. CryptoSlate (2025-11) — BTC-DXY returns vs levels: ~ 0.04 same-day, ~ -0.20 best 33-day lead, vs -0.58 spurious level correlation.

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